

C1 implements C2 extends I1

Program Code:

```
package core_java;

interface I1 {
    void calculateBill();
}

class C2 implements I1 {

    public void calculateBill() {
        System.out.println("Total bill calculated");
    }

    void generateReceipt() {
        System.out.println("Receipt generated");
    }
}

class C1 extends C2 {

    void makePayment() {
        System.out.println("Payment completed successfully");
    }
}

public class C1C2I1 {
    public static void main(String[] args) {

        C1 obj = new C1();

        obj.calculateBill();
        obj.generateReceipt();
        obj.makePayment();
    }
}
```

Explanation:

This program demonstrates an interface and inheritance relationship. Interface I1 declares calculateBill(). Class C2 implements I1 and provides the implementation. Class C1 extends C2, so its object can access the implemented interface method and the methods inherited from C2, along with its own method.

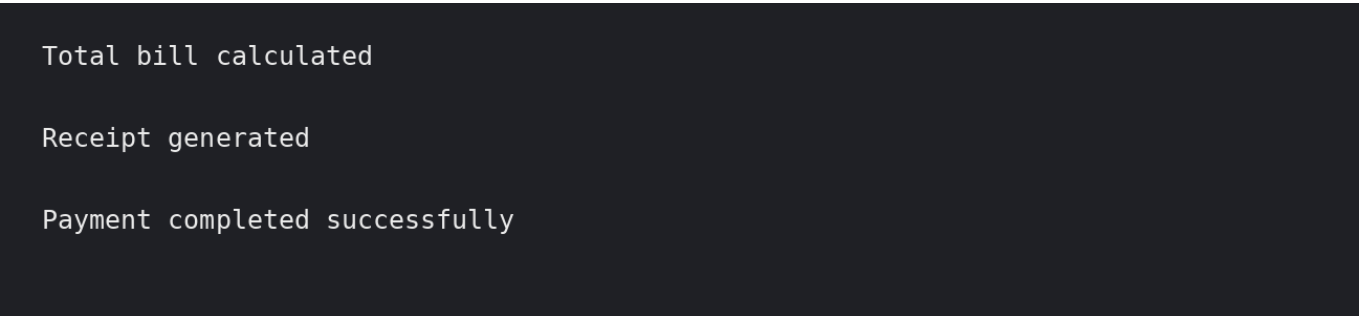
Compilation and Execution Commands:

Save the file as: C1C2I1.java

Compile:
javac -d . C1C2I1.java

Run:
java core_java.C1C2I1

Output Screenshot:



```
Total bill calculated

Receipt generated

Payment completed successfully
```

Verification:

The output confirms that the C1 object accesses calculateBill() through the interface implementation in C2, inherits generateReceipt() from C2, and executes its own makePayment() method.

Result:

Thus, the program was successfully executed and the required relationship involving C1, C2 and I1 was verified successfully.